

Javascript

Dom events, dom methods

```
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>DOM Events</title>
    <style>
      .active {
        background-color: #333;
        color: aqua;
      }

      .nonactive {
        background-color: red;
        color: white;
      }
      #container {
        width: 100px;
        height: 100px;
        background-color: #333;
      }
    </style>
  </head>
  <body>
    <h2 id="h2Element" class="nonactive">DOM</h2>

    <button id="button">Toggle class</button>
    <div id="container"></div>

    <form id="myForm">
      <input type="text" id="inp-box" placeholder="Enter name" />
      <button type="submit">Submit</button>
    </form>

    <script>

      document.getElementById("myForm").onsubmit = function(e){
        e.preventDefault();
        alert("Form submitted");
      }

      document.getElementById("inp-box").oninput = function(e){
        console.log("oninput method:",this.value);
      }
    </script>
  </body>
</html>
```

```

    }
    document.getElementById("inp-box").onchange = function(e){
        console.log("onchange method :",this.value);
    }

    window.onload = ()=>{
        console.log("Page loaded successfully");
    }
    window.onresize = ()=>{
        console.log("Resize");
    }

    window.onscroll = ()=>{
        console.log("Scrolling");
    }
    // let h2Element = document.getElementById("h2Element");

    // let button = (document.getElementById("button").onclick = () => {
    //     alert("clicked");
    // });

    // document.getElementById("container").onmouseover = function () {
    //     this.style.backgroundColor = "Blue";
    // };

    // document.getElementById("container").onmouseout = function () {
    //     this.style.backgroundColor = "#333";
    // };

    // document.addEventListener("keydown", function (event) {
    //     if (event.key == "Enter") {
    //         console.log("User submitted form successfully");
    //     }
    //     console.log("Key pressed: ", event.key);
    // });

    // document.addEventListener("keyup", function (event) {
    //     console.log("Key UP: ", event.key);
    // });

    // button.addEventListener("click", () => {
    //     h2Element.classList.toggle("nonactive");
    // });

```

```
// h2Element.classList.add("active");  
// h2Element.classList.remove("nonactive");  
</script>  
</body>  
</html>
```

Theory

1. Form Events

- **onsubmit** → triggered when a form is submitted.
- `e.preventDefault()` prevents default behavior (like page reload).

2. Input Events

- **oninput** → fires whenever the value changes (real-time typing).
- **onchange** → fires when input loses focus and value has changed.

3. Window Events

- **onload** → fires when the page finishes loading.
- **onresize** → fires when the window is resized.
- **onscroll** → fires when the page is scrolled.

4. Mouse Events

- **onmouseover** → triggered when mouse enters element.
- **onmouseout** → triggered when mouse leaves element.

5. Keyboard Events

- **keydown** → fires when a key is pressed down.
- **keyup** → fires when a key is released.

6. Class Manipulation

- `element.classList.add("className")` → add class.
- `element.classList.remove("className")` → remove class.
- `element.classList.toggle("className")` → toggle class.

Coding Round Questions

- Form events → Explain difference between `onsubmit` and `preventDefault()`.
- Input events → What is the difference between `oninput` and `onchange`?
- Window events → Demonstrate `onload`, `onresize`, and `onscroll`.
- Mouse events → Show how `onmouseover` and `onmouseout` work.
- Keyboard events → Explain `keydown` vs `keyup`.
- Class manipulation → Show how to toggle classes dynamically.

⚡ Machine Coding Round Challenge

Problem: Interactive Form & Box

Requirements:

1. Create a form with input and submit button.
2. On submit, show an alert with entered value.
3. Add a box that changes color on hover.
4. Toggle a heading's class on button click.
5. Log key presses in the console.